Jayanth (Jay) Palla

Email: jaypalla22@gmail.com Mob: 484-239-3465 LinkedIn: www.linkedin.com/in/jaypalla
Gitrepo: github.com/TigerShock4982

Summary

Entry-level Software Engineer with software development and observability experience building telemetry, observability, and large-scale data processing systems using Python, SQL, Databricks, Apache Spark, and modern lakehouse architectures. Experienced developing backend ingestion pipelines, transforming structured and semi-structured data, implementing real-time monitoring dashboards, and supporting observability using Splunk, Grafana, Prometheus, and Cribl. Strong foundation in ETL workflows, API based ingestion, modelling, cloud native infrastructure, and CI/CD practices. **Experience**

Software Engineer, Keylent Inc, Delaware *Dec 2025 – Present* Client: Barclaycard US

- Working on a financial transaction monitoring platform for payment activity, tracks system health, and provides observability across backend services using logs, metrics, traces, and alerts.
- Designed dashboards to monitor real time transaction volume, latency, error rates, fraud-rule triggers, and service reliability of Kafka messages for a banking/payment system.
- Helped onboarding application and infrastructure data into Splunk by using Cribl to transform, filter, and route log streams through custom observability pipelines.

Tech Stack: Grafana, Splunk, Prometheus, Cribl, Python structured logging, JSON logs, Oracle, SQL Server

Software Engineering Intern, Keylent Inc, Delaware *May 2025 - Aug 2025*

- Automation of manual tasks using playbooks to schedule updates and execute scripts in parallel on all servers in a cluster. This improved efficiency brought down the downtime, reduced human error, and enabled blue/green deployments.
- Created Splunk reports and Grafana dashboards that help in bringing observability to the systems that are supported for some critical business functions.

Tech Stack: Grafana, Splunk, Python, SQL Server

Capstone, Penn State University *Jan 2026 - May 2026*

- Designed and developed a scalable telemetry and analytics platform using Databricks, Apache Spark, and Delta Lake concepts to process real IoT sensor information from a smart hydroponic vertical farm setup
- Built ETL pipelines to ingest structured JSON streams including environmental, water quality, and device health metrics into analytics-ready datasets
- Implemented Bronze/Silver/Gold processing layers to support raw ingestion, cleaned transformations, and dashboard-ready aggregate datasets
- Developed workflows that support schema validation, incremental processing, error handling, and reusable components
- Created real-time monitoring dashboards and observability tooling to track pipeline health and system metrics
- Applied Spark based transformations to optimize querying, aggregation, and historical telemetry analysis across large time-series datasets, including scalability for multiple farm setups
- Focused on scalable data engineering patterns including distributed processing, lakehouse design principles, observability, enrichment, and enablement for future AI/ML use cases
- Received First Place People's Choice Award and First Place Video Award for this project from Penn State

Tech Stack: Databricks, Apache Spark, PySpark, Delta Lake, SQL, Python, FastAPI, Grafana, Splunk, Git

Research Assistant, Penn State University *May 2024 – May 2025*

- Worked in the Penn State College of Science department on research of graph structures
- Focused on edge connectivity *K* of simple connected graph structures to develop and prove conjectures regarding networks, with applications across industries
- Developed high level programming using Python to efficiently test the conjectures made in the research process using Pandas DataFrames to store and manipulate vertex-edge relationships
- Developed a real-time telemetry monitoring platform using embedded sensors, microcontroller data transmission, backend APIs, and dashboard visualization.
- Built backend services to ingest, validate, and store telemetry data including temperature, humidity, pressure, light, water quality, and device status readings.
- Implemented live monitoring features to track sensor trends, device sequence numbers, data freshness, and alert conditions for abnormal system behavior.
- Designed a dashboard to display current sensor values, historical trends, and system status indicators, improving visibility into hardware and software performance.

Tech Stack: NumPy, Python, Pandas, NetworkX, Supabase, TypeScript, FastAPI

Skills and Proficiencies

- Programming languages **Python**, **Java**
- HTML5, CSS, Node.js, **Angular JS, ReactJS**, **Spring Boot**, **Pandas**, **NumPy**
- **Data Ingestion** and **ETL Pipeline Development**
- Observability tools **Grafana, Splunk**, **Prometheus, AppDynamics, Cribl**
- **Databricks, Apache Spark, Delta Lake** architecture
- **Lakehouse Performance Optimization**
- **SQL Server** and **Oracle RDBMS** including extraction, loading, querying, transportation, and analysis
- **Data Governance** and **Metadata** experience
- Understanding of **NoSQL** databases like **MongoDB**
- Experience with Cloud Technologies (**Azure, AWS, Google**)
- **Cribl Certified User** accreditation
- **GUI Development** (JavaFX Scene Builder)
- Familiar with **Agile** methodologies (Scrum, Kanban)
- MS Office Tools (**Word / Excel / PowerPoint / Visio**)
- Strong logical reasoning, perseverance and analytical skills

Education

- **Penn State University, University Park** - B.S **Computer Engineering**

Work Authorization: No sponsorship required